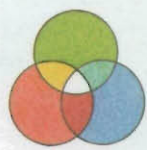


COLOUR

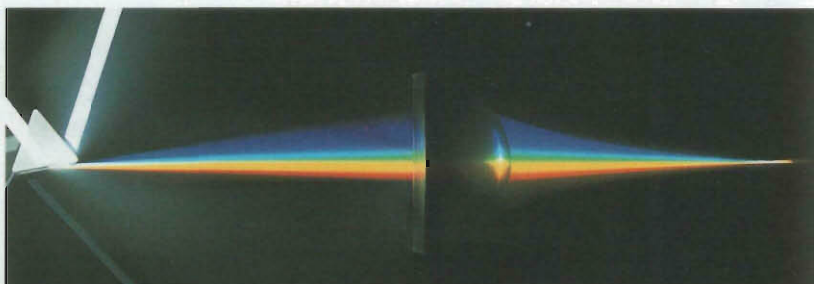


A WORLD WITHOUT COLOUR would be dull and uninspiring. Colour is a form of light. Light is made up of electromagnetic waves of

varying lengths. The human eye detects these different wavelengths and sees them as different colours. White light – like that from the Sun – is a mixture of all the different wavelengths. Objects look coloured because they give out or reflect only certain wavelengths of light.

White light spectrum

Passing white light through a transparent triangular block called a prism separates out the different wavelengths of light. The prism refracts (bends) each wavelength by a different amount, forming a band of colours called a white light spectrum, or a visible spectrum. The seven main colours are red, orange, yellow, green, blue, indigo, and violet. Red has the longest wavelength and violet the shortest. Here, a convex lens combines the colours back into white light.



A rainbow at dawn

Rainbow

If it rains on a sunny day, you may well see a rainbow if you stand with your back to the Sun. A rainbow is a curved white light spectrum that forms when light is reflected and refracted by raindrops in the sky.

How a rainbow forms

When white sunlight passes through a raindrop, the raindrop acts like a tiny prism. The raindrop refracts the light and splits it up into its separate colours. The colours fan out and emerge as a spectrum. A rainbow is made up of spectra from millions of raindrops.



Colour and temperature

Objects at room temperature emit (give out) electromagnetic waves, but these waves are too long for human eyes to see. Heating an object, such as this steel bar, gives the waves it emits more energy and makes them shorter. The waves eventually become short enough to be seen, and the bar begins to glow. As the bar's temperature rises, it glows with different colours.



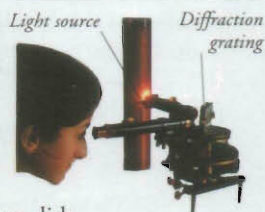
Steel bar at 630°C (1,170°F)



Steel bar at 1,530°C (2,790°F)

Spectroscope

An instrument called a spectroscope is used to analyze the light given out by hot substances. Inside the spectroscope, a prism or diffraction grating (a glass slide scored with fine lines) splits light from a glowing substance into its component wavelengths.



Emission spectrum

Each chemical element gives out a unique range of light wavelengths when heated. Seen through a spectroscope, these wavelengths appear as a set of bright lines on a dark background. This is the element's emission spectrum. A compound's emission spectrum is a combination of spectra from the elements that make up the compound.



Emission spectrum of a sodium flame

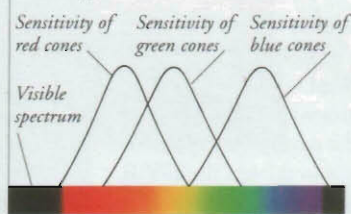
Sodium flame

Cone cells

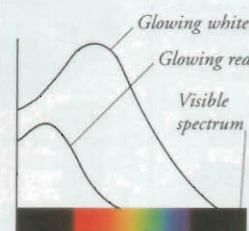
At the back of the eye there are special cells called cones that enable humans to see colours. There are three types of cone, called red, green, and blue cones. Each type of cone is sensitive to a different range of light wavelengths. White light stimulates all three types of cone.



Cone cells



Sensitivity of cone cells in the human eye



Hot stars

The colour of a star gives a clue to its age. To the naked eye, most stars look white, but their true colours can be seen using a telescope. Young stars are hot and glow with white light. Older stars are relatively cool and glow red or orange.

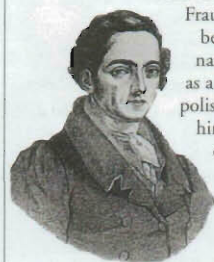
Red hot and white hot
As the steel bar gets hotter, it emits more and more of the visible spectrum. At about 630°C (1,170°F), it is "red hot" and emits light from the red end of the spectrum. At about 1,530°C (2,790°F), the "white hot" bar emits the entire white light spectrum.



A cluster of young stars

Joseph von Fraunhofer

The German physicist Joseph von Fraunhofer (1787–1826) became interested in the nature of light while training as a mirror maker and lens polisher. His training enabled him to make spectroscopes of great precision. From 1814–17, he used them to make the first scientific study of the Sun's emission spectrum.



Munsell colour system

Describing colours exactly using words alone is not easy. To avoid confusion, manufacturing industries use standard colour-identification systems. The Munsell system is used to specify colours for dyes and pigments. It defines a colour by its value (brightness), its chroma (strength), and its hue (position in the spectrum).



Colour matching systems

Graphic designers use swatches of colour cards to match the colours in their work with those available from printers. The designer supplies the printer with the reference number of the colour, so the printer knows exactly what is wanted.

Each colour has a reference number.

